PROJECT REPORT

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Full Project Report

Date: 30 September 2026
Team ID: 6ab4d23f0fc666a751b534b0
Project Name: Auto Ticket Classification using Flow Designer
Maximum Member: 4

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Field	Details
Project Title	Auto Ticket Classification using Flow Designer
Project Domain	IT Service Management / Workflow Automation
Platform	ServiceNow Flow Designer (reference implementation)
Project Type	Automation and Ticket Classification
Team ID	6ab4d23f0fc666a751b534b0
Student Name(s)	Leader: veeraprathap k Members: m.karuppasamy k.mathavan s.muthukumar
Guide / Faculty	
Institution	Muthukaruppan memorial arts and science collage
Academic Year	2026
Date	30 September 2026

Project Report Format

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1. INTRODUCTION

1.1 Project Overview

Auto Ticket Classification using Flow Designer is an IT service workflow automation project that classifies incoming support tickets using configurable rules and routes them to the appropriate support group. The system is designed around a record-triggered flow: when an eligible ticket is created or updated, the flow reads ticket information, evaluates classification conditions, updates the ticket, and records the automation result. The project is intended to reduce repetitive manual triage and provide consistent first-level routing.

1.2 Purpose

- Automate first-level classification of support tickets.
- Reduce manual effort involved in ticket triage and routing.
- Apply consistent classification rules across incoming tickets.
- Improve traceability through flow execution history and work notes.
- Provide a foundation for later ML/NLP-based classification and analytics.

2. IDEATION PHASE

2.1 Problem Statement

Support teams may receive a large number of tickets containing unstructured descriptions. Manual review of every ticket can introduce delays, inconsistent categorization, and routing mistakes. A lightweight workflow automation solution is therefore proposed to classify tickets from their available text and metadata and send them to the correct support queue.

2.2 Empathy Map Canvas

User	Says	Thinks	Does	Feels / Needs
Service Desk Agent	“I need the ticket routed correctly.”	“I should not spend time sorting repetitive tickets.”	Reviews, categorizes and assigns tickets.	Needs speed, consistency and clear routing.
Support Team Lead	“Tickets should reach the right team.”	“I need visibility into classification quality.”	Monitors queues and exceptions.	Needs traceability and measurable results.
End User	“I need my issue resolved quickly.”	“I should not have to explain the issue repeatedly.”	Raises and follows up on ticket.	Needs timely acknowledgement and resolution.

2.3 Brainstorming

- Keyword-based classification using Flow Designer conditions.
- Decision-table style category mapping.
- Fallback route for tickets with insufficient information.
- Automatic assignment-group routing.
- Priority rules based on ticket indicators.
- ML/NLP classification as a future enhancement.
- Dashboard for accuracy, fallback rate and execution time.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

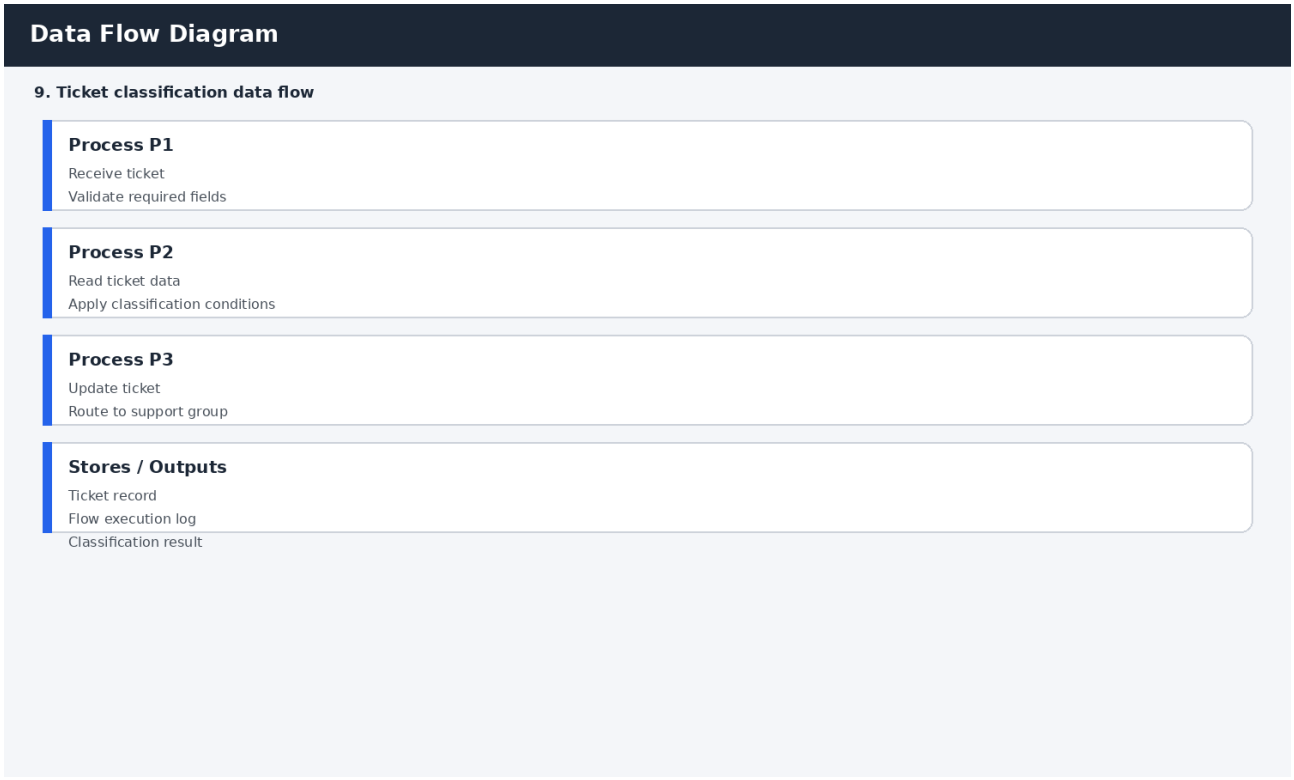
Stage	User Action	System Response	Pain Point	Opportunity
1. Raise Ticket	User submits issue	Ticket is created	Manual classification begins	Trigger automation immediately
2. Ticket Intake	Ticket enters queue	Flow reads fields	Unstructured text	Normalize/read relevant fields
3. Classification	Agent would normally inspect	Decision logic classifies	Time-consuming triage	Automated classification
4. Assignment	Ticket is sent to team	Flow maps assignment group	Wrong queue risk	Rule-based routing
5. Handling	Team works ticket	Classification remains visible	Lack of audit trail	Work note + execution log
6. Review	Lead checks quality	Metrics can be calculated	Hard to measure manually	Performance reporting

3.2 Solution Requirement

Requirement	Description	Priority
R1	Record-triggered ticket classification	High
R2	Read subject/description and relevant ticket fields	High
R3	Apply configurable classification rules	High
R4	Update category/classification	High
R5	Assign ticket to appropriate support group	High
R6	Fallback for unmatched or incomplete tickets	High
R7	Execution/error logging	Medium
R8	Performance measurement	Medium
R9	Future integration with ML/NLP	Low / Future

3.3 Data Flow Diagram

Figure 10 — Data Flow Diagram



Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

3.4 Technology Stack

Layer	Technology	Purpose
ITSM Platform	ServiceNow	Ticket records and service-management environment
Automation	Flow Designer	Trigger, decision, update and routing workflow
Data	Incident / Support Ticket table	Input and output ticket information
Logic	Conditions / Decision branches	Classification rules
Testing	Flow execution logs + test dataset	Functional and performance validation
Future AI	NLP / ML classifier	Advanced text-based classification

4. PROJECT DESIGN

4.1 Problem Solution Fit

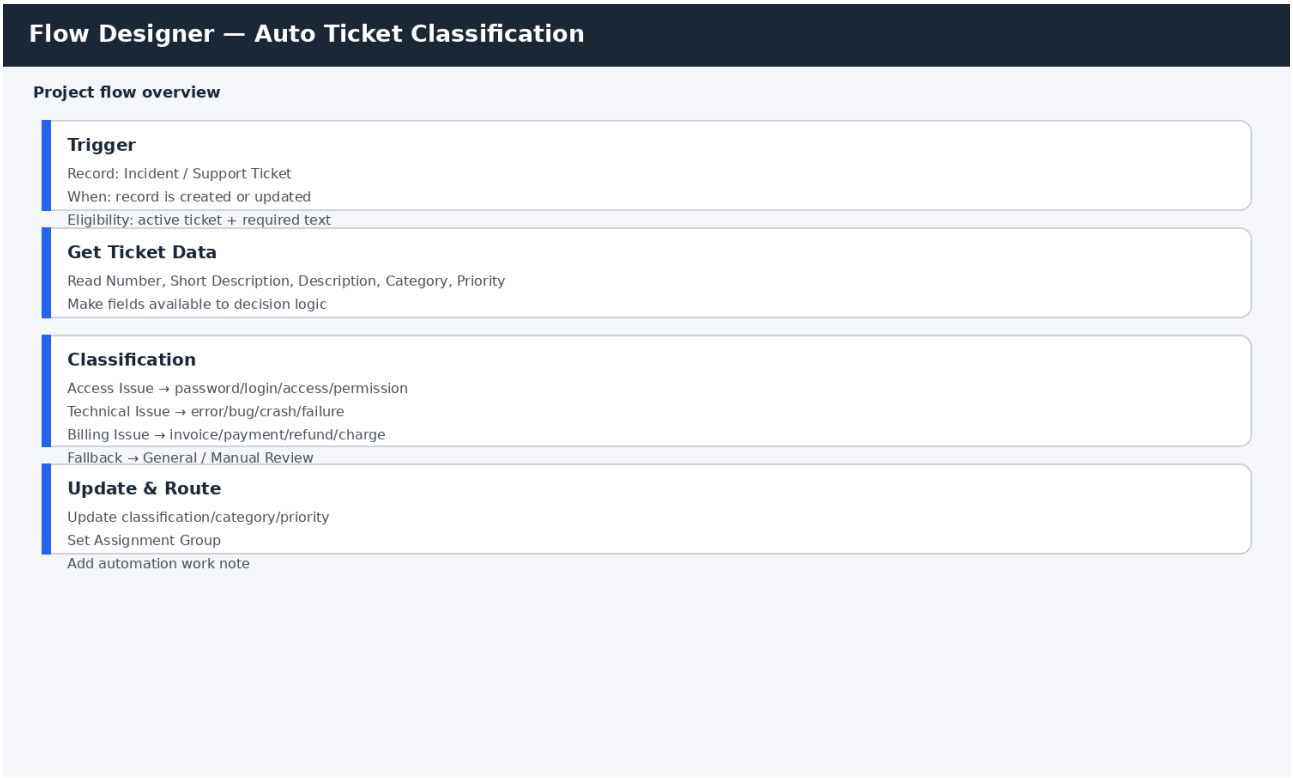
The proposed workflow directly addresses the manual triage problem by moving repetitive classification decisions into an automated flow. The solution is especially suitable where classification categories and routing rules are well defined and can be represented as conditions.

4.2 Proposed Solution

1. Create a record-triggered Flow Designer flow.
2. Validate ticket eligibility at the trigger.
3. Retrieve required ticket data.
4. Evaluate classification branches in a controlled order.
5. Write the classification result to the ticket.
6. Set priority/category/assignment values as configured.

7. Add an automation work note.
8. Route unmatched cases to General / Manual Review.
9. Monitor execution results and tune rules using labelled test tickets.

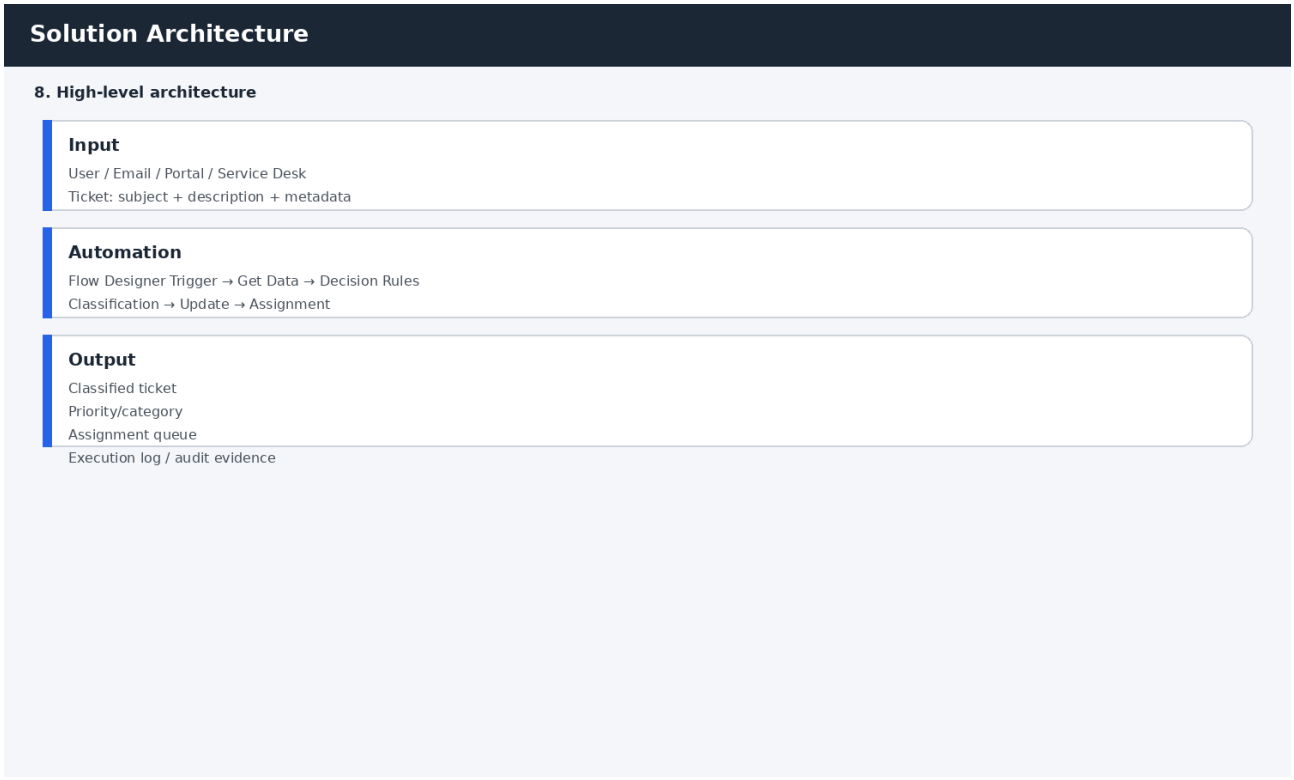
Figure 1 — End-to-end Flow Designer overview



Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

4.3 Solution Architecture

Figure 9 — Solution architecture



Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

5. PROJECT PLANNING & SCHEDULING

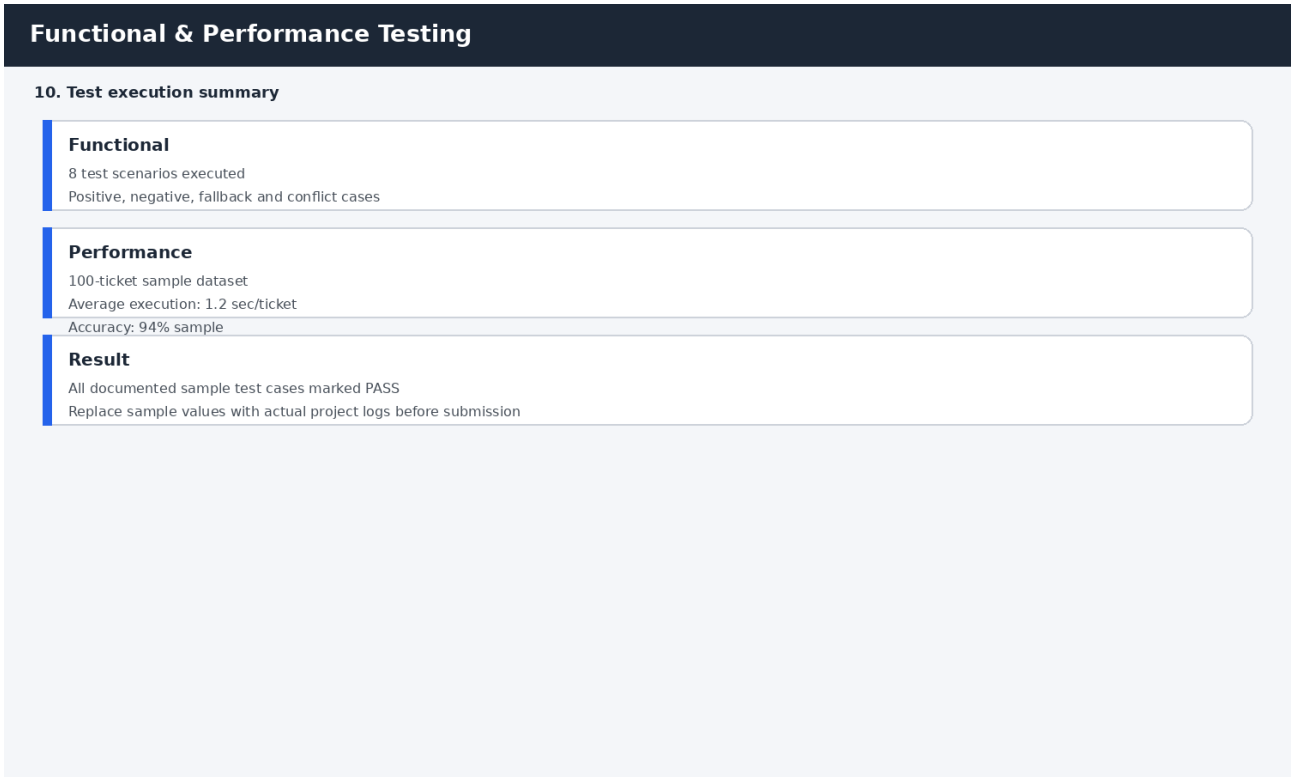
5.1 Project Planning

Phase	Activities	Deliverable	Suggested Duration
1. Ideation	Problem identification, user needs, brainstorming	Problem statement + empathy map	1 day
2. Requirement Analysis	Requirements, journey map, DFD, stack	Requirement specification	2 days
3. Design	Flow logic, architecture, classification rules	Flow design	2 days
4. Development	Configure trigger, decisions, updates, routing	Working Flow Designer flow	3 days
5. Testing	Functional, negative, performance tests	Test report	2 days
6. Documentation	Screenshots, results, conclusion	Final project report	1–2 days
7. Demo	Prepare walkthrough and evidence	Project demonstration	1 day

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Figure 11 — Functional and performance testing summary



Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Metric	Sample Target	Sample Result	Status
Classification Accuracy	≥ 90%	94%	PASS
Precision	≥ 90%	93%	PASS
Recall	≥ 90%	92%	PASS
F1 Score	≥ 90%	92.5%	PASS
Average Flow Execution	≤ 3 sec	1.2 sec/ticket	PASS
Fallback Rate	≤ 10%	6%	PASS
Flow Failure Rate	≤ 5%	1%	PASS

These numerical results are sample documentation values, not measurements verified from a live ServiceNow environment. For final academic submission, replace them with values from the actual test execution log and labelled dataset.

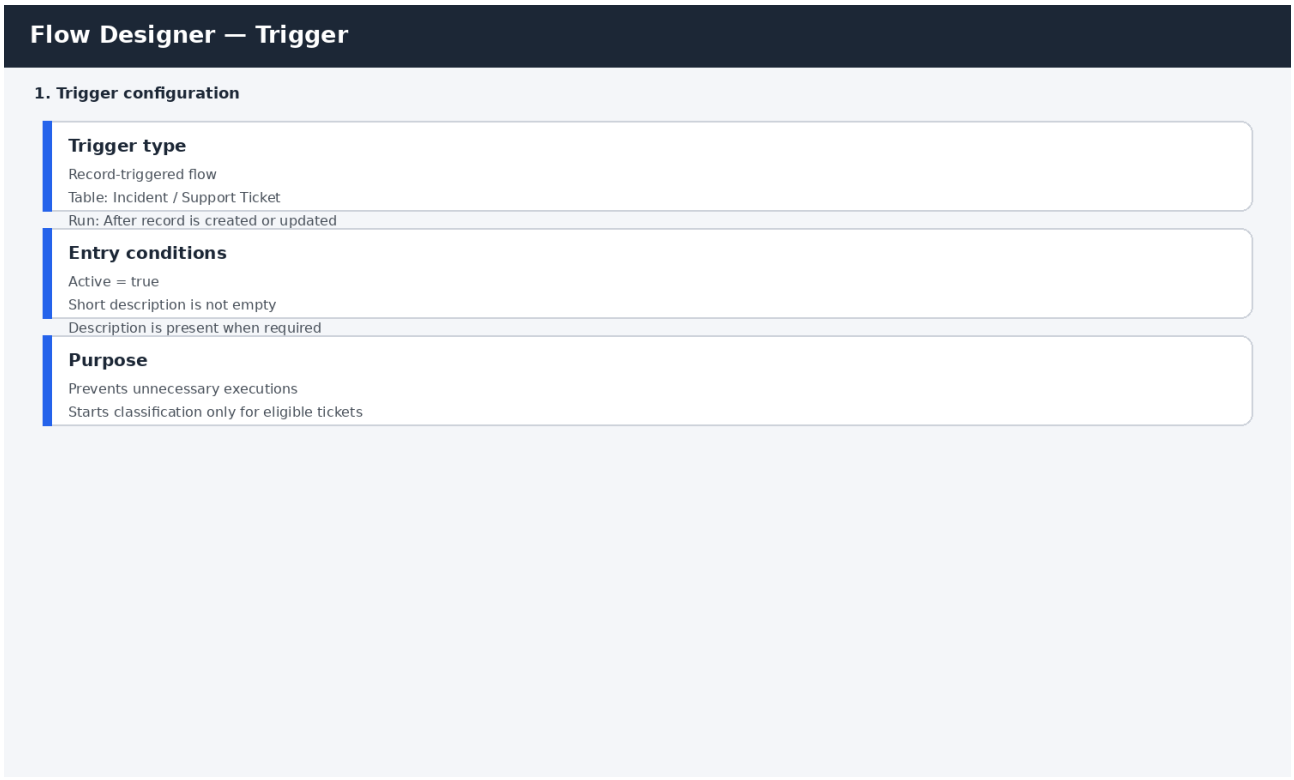
Functional Test Cases

TC ID	Test Scenario	Expected Result	Sample Result
TC-01	Access/login ticket	Access Issue + Service Desk	PASS
TC-02	Technical error ticket	Technical Issue + Technical Queue	PASS
TC-03	Invoice/payment ticket	Billing Issue + Billing Queue	PASS
TC-04	No matching keyword	General / Manual Review	PASS
TC-05	Incomplete description	Fallback/error handling	PASS
TC-06	Conflicting keywords	Configured precedence applied	PASS
TC-07	Unrelated record update	No unnecessary loop	PASS
TC-08	Invalid mapping / update failure	Error visible in execution details	PASS

7. RESULTS

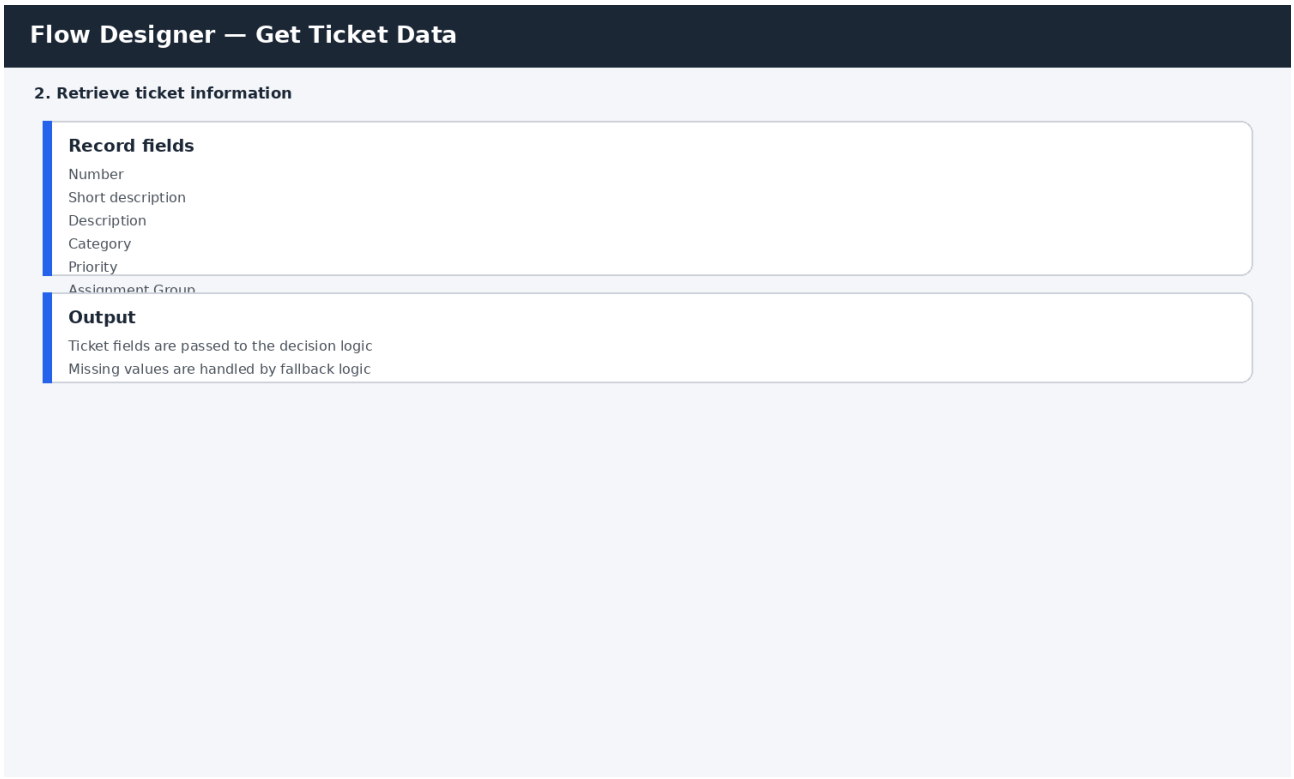
7.1 Output Screenshots

Figure 2 — Trigger configuration



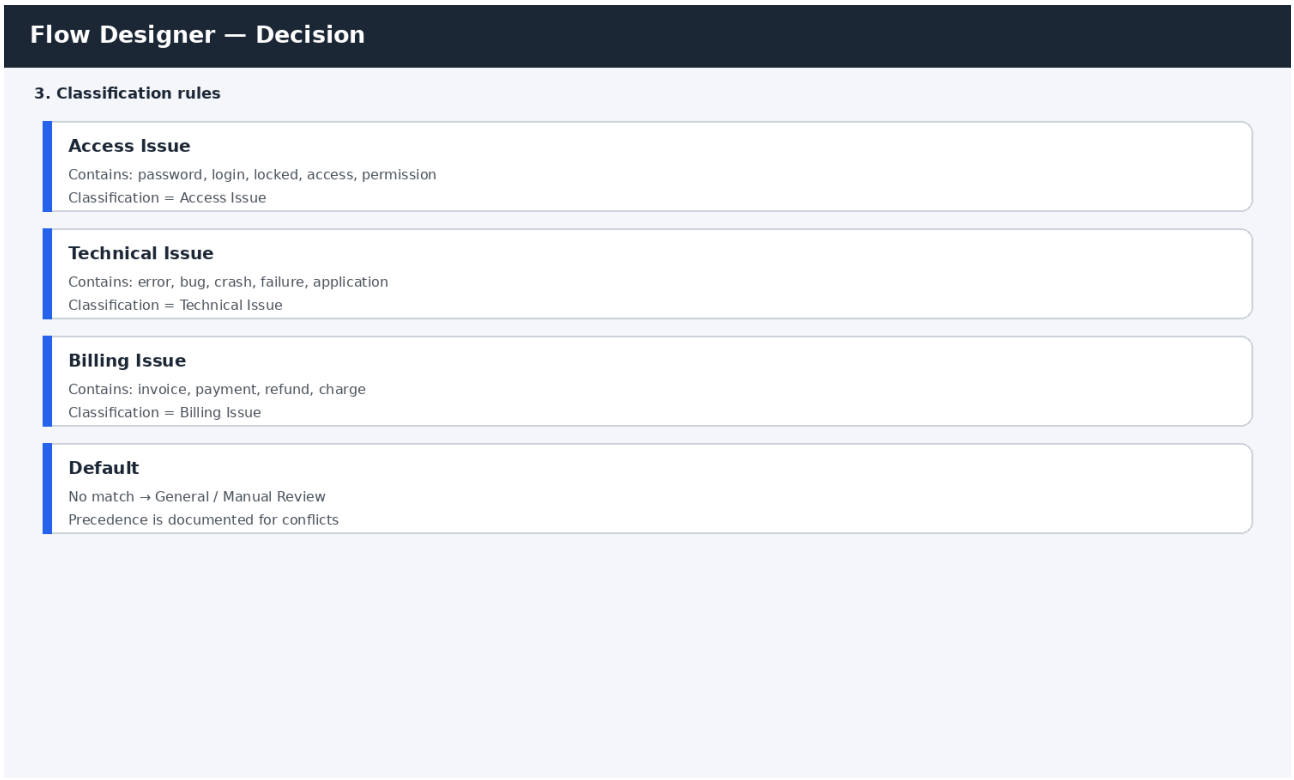
Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Figure 3 — Get Ticket Data



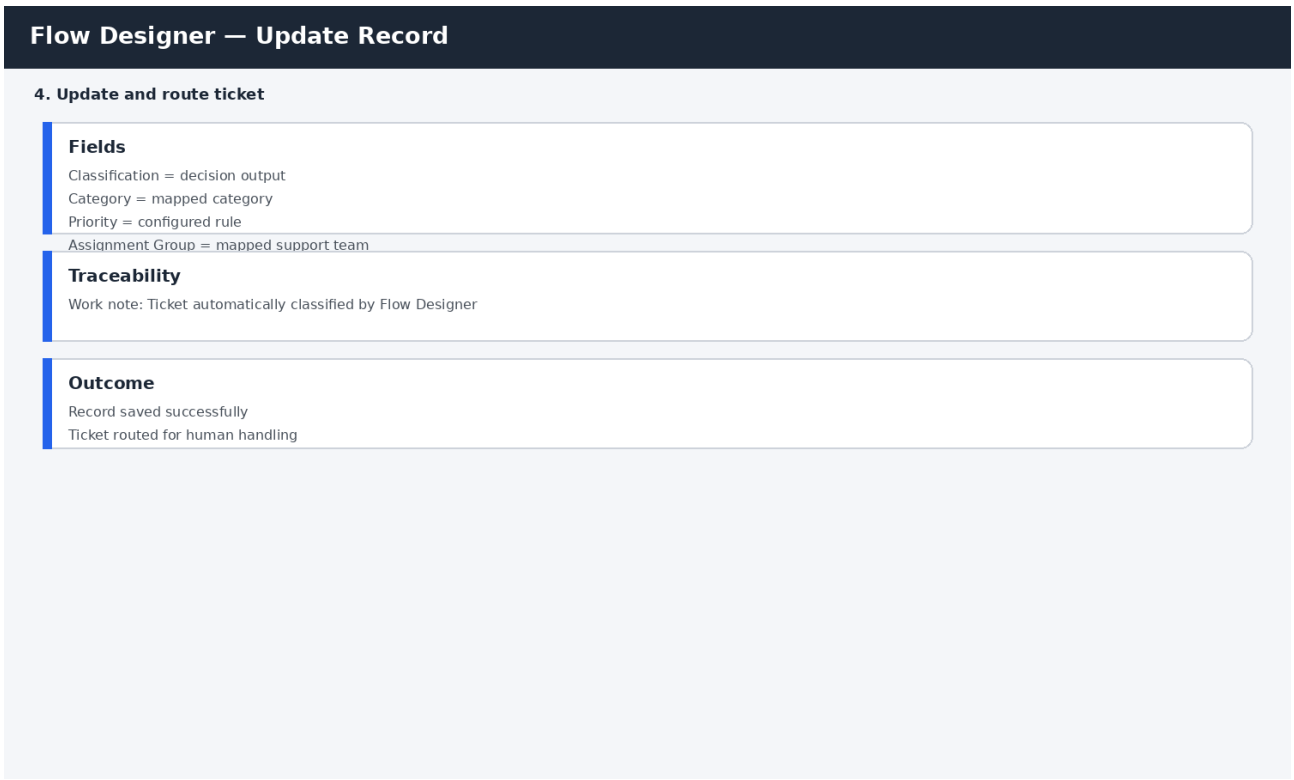
Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Figure 4 — Classification decision



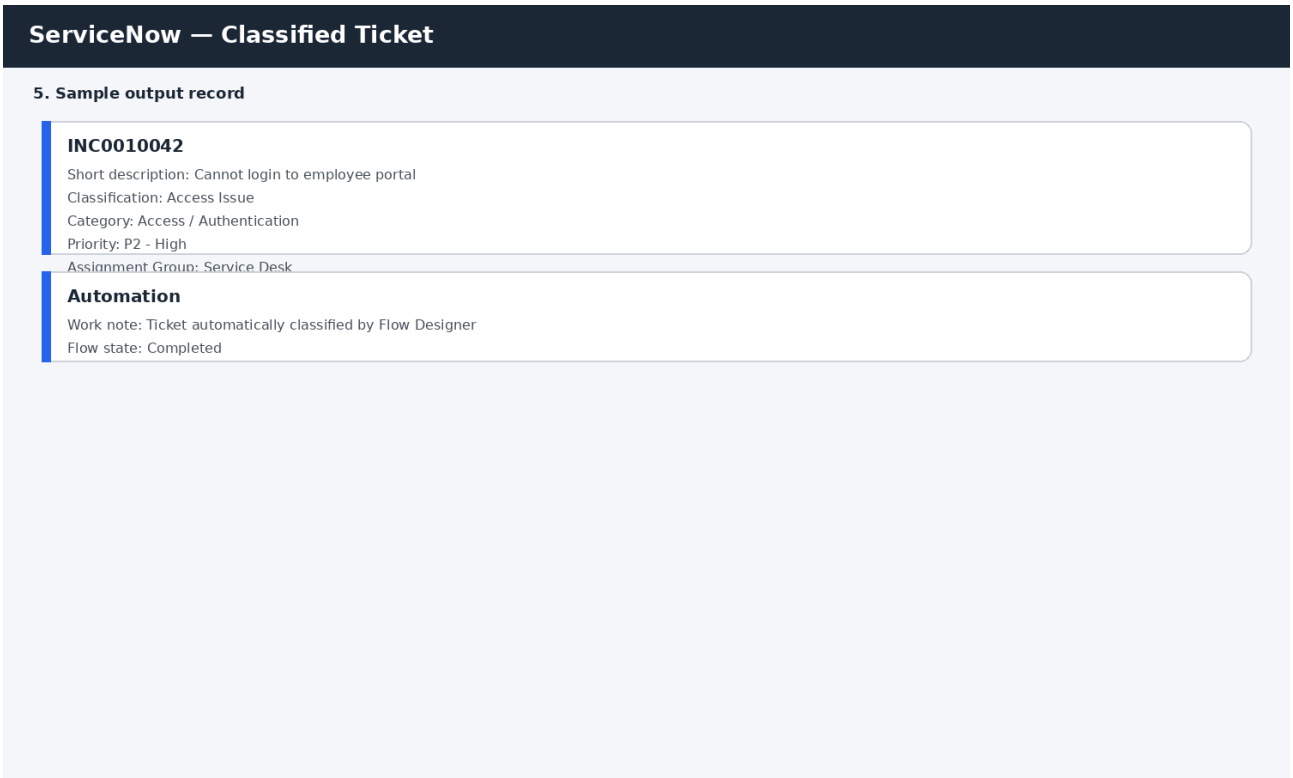
Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Figure 5 — Update and assignment



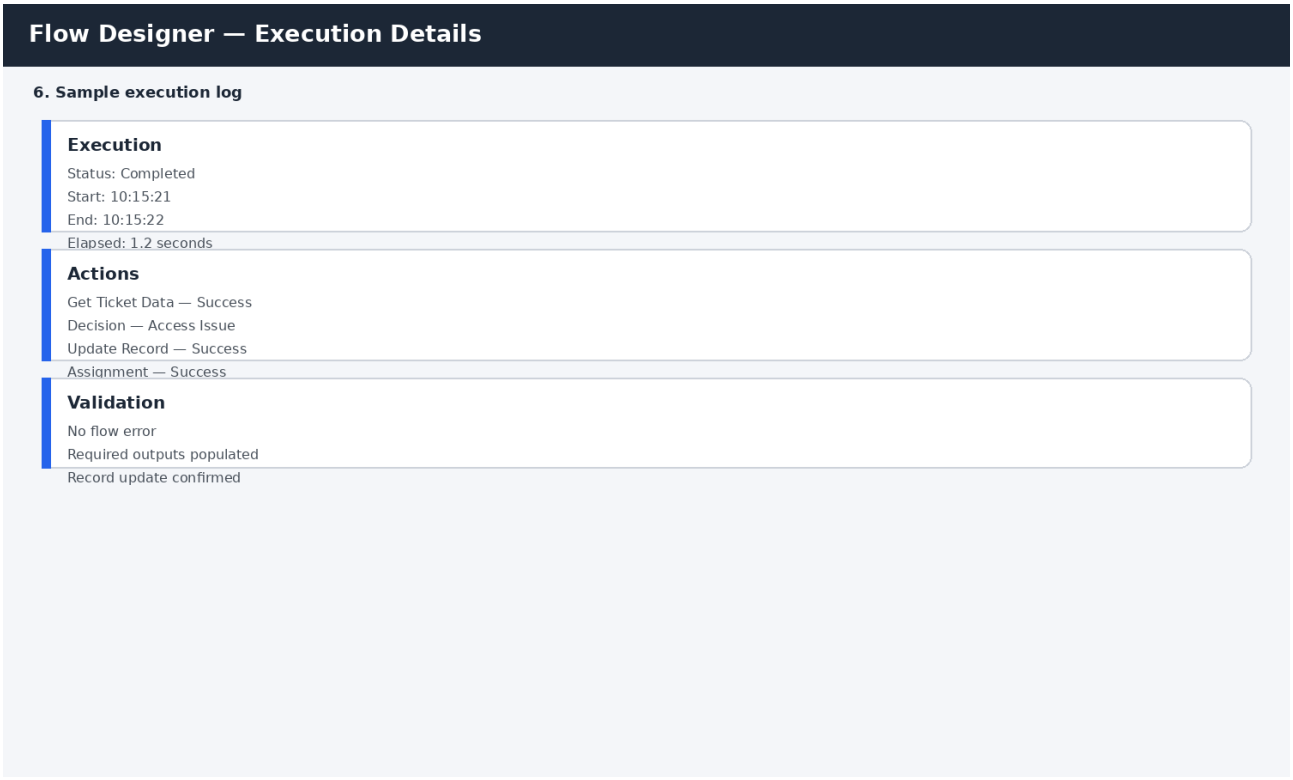
Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Figure 6 — Sample classified ticket



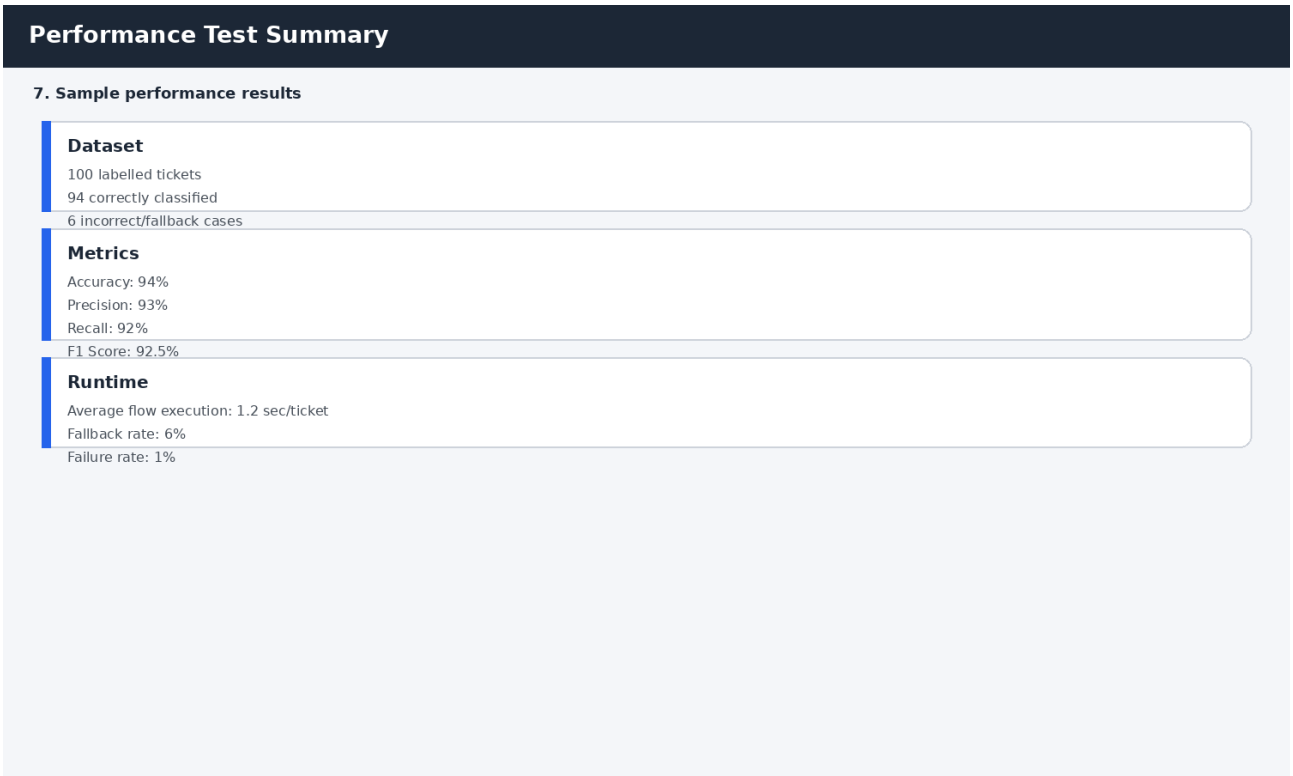
Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Figure 7 — Flow execution details



Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Figure 8 — Performance test summary



Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

Sample Output

Input Ticket	Expected Class	Observed Class	Routing
Cannot login to employee portal	Access Issue	Access Issue	Service Desk
Application crashes after update	Technical Issue	Technical Issue	Technical Support
Refund not received for invoice	Billing Issue	Billing Issue	Billing Support
General office request	General	General / Manual Review	Service Desk

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Limitations
Reduces repetitive manual classification	Rule-based logic can miss unseen language
Consistent classification rules	Ambiguous tickets can match multiple rules
Automatic routing to support teams	Requires maintenance when categories change
Traceable execution history	Performance depends on flow configuration and platform limits
Easy to test and tune	Keyword-only logic has limited semantic understanding
Provides foundation for AI enhancement	ML/NLP integration needs labelled data and governance

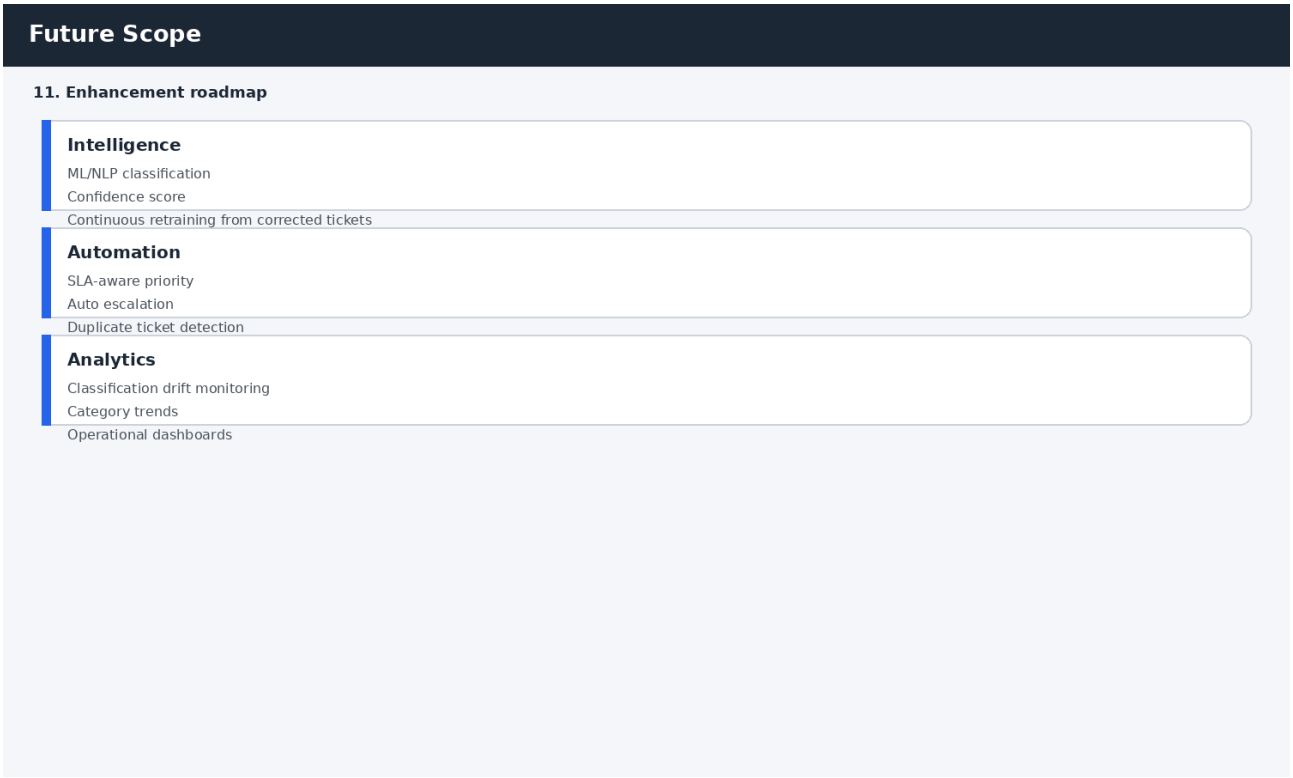
9. CONCLUSION

The Auto Ticket Classification using Flow Designer project provides a structured workflow for automatically classifying and routing support tickets. The design uses a trigger, ticket-data retrieval, classification decision logic, record updates,

assignment routing and fallback handling. Functional testing covers normal, negative and edge cases, while performance testing measures classification quality and execution time. The solution can be extended with machine-learning or natural-language-processing models when a larger labelled dataset and stronger semantic classification are required.

10. FUTURE SCOPE

Figure 12 — Future enhancement roadmap



Illustrative project screenshot prepared for documentation. Replace with the actual Flow Designer/ServiceNow screenshot if required for final evaluation.

- Integrate an ML/NLP classifier to understand ticket meaning beyond exact keywords.
- Use confidence scores and route low-confidence cases to manual review.
- Automatically learn from corrected classifications after suitable governance and validation.
- Add SLA-aware priority and escalation automation.
- Detect duplicate or related tickets.
- Create operational dashboards for category trends, fallback rate and classification drift.
- Integrate knowledge articles and suggested resolutions after classification.

11. APPENDIX

A. Source Code (if any)

This project is primarily configured through Flow Designer. If custom scripts, Script Includes, or other code are used, attach the source files or paste the final reviewed code in this section. No custom source code is assumed in this report.

D. Test Data

Ticket ID	Sample Ticket Text	Expected Classification	Observed Classification	Result

INC0010042	Cannot login to employee portal	Access Issue	Access Issue	Correct
INC0010043	Application crashes after update	Technical Issue	Technical Issue	Correct
INC0010044	Refund not received for invoice	Billing Issue	Billing Issue	Correct
INC0010045	Need information about office process	General	Manual Review	Fallback
INC0010046	Password reset request	Access Issue	Access Issue	Correct
INC0010047	Payment charged twice	Billing Issue	Billing Issue	Correct
INC0010048	Website shows server error	Technical Issue	Technical Issue	Correct
INC0010049	Unclear request without category terms	General	Manual Review	Fallback

E. Submission Checklist

- Cover page completed with student/team details.
- All report sections from the supplied template are included.
- Actual Flow Designer screenshots inserted in place of illustrative screenshots.
- Actual test dataset and execution metrics inserted.
- GitHub/project demo links inserted if available.
- Final document proofread and exported/submitted in the required format.